



Liquid fuels demand and supply model

Week 1 analytical pack

South Africa | Petrol and diesel

6 October 2026

Internal working discussion

Photo source: Royal Vopak | Terminal Durban | vopak.com



Resolve the SA fuel outlook, then test Vopak's accessible market

Situation → Complication → Resolution | evidence, actions and delivery roadmap

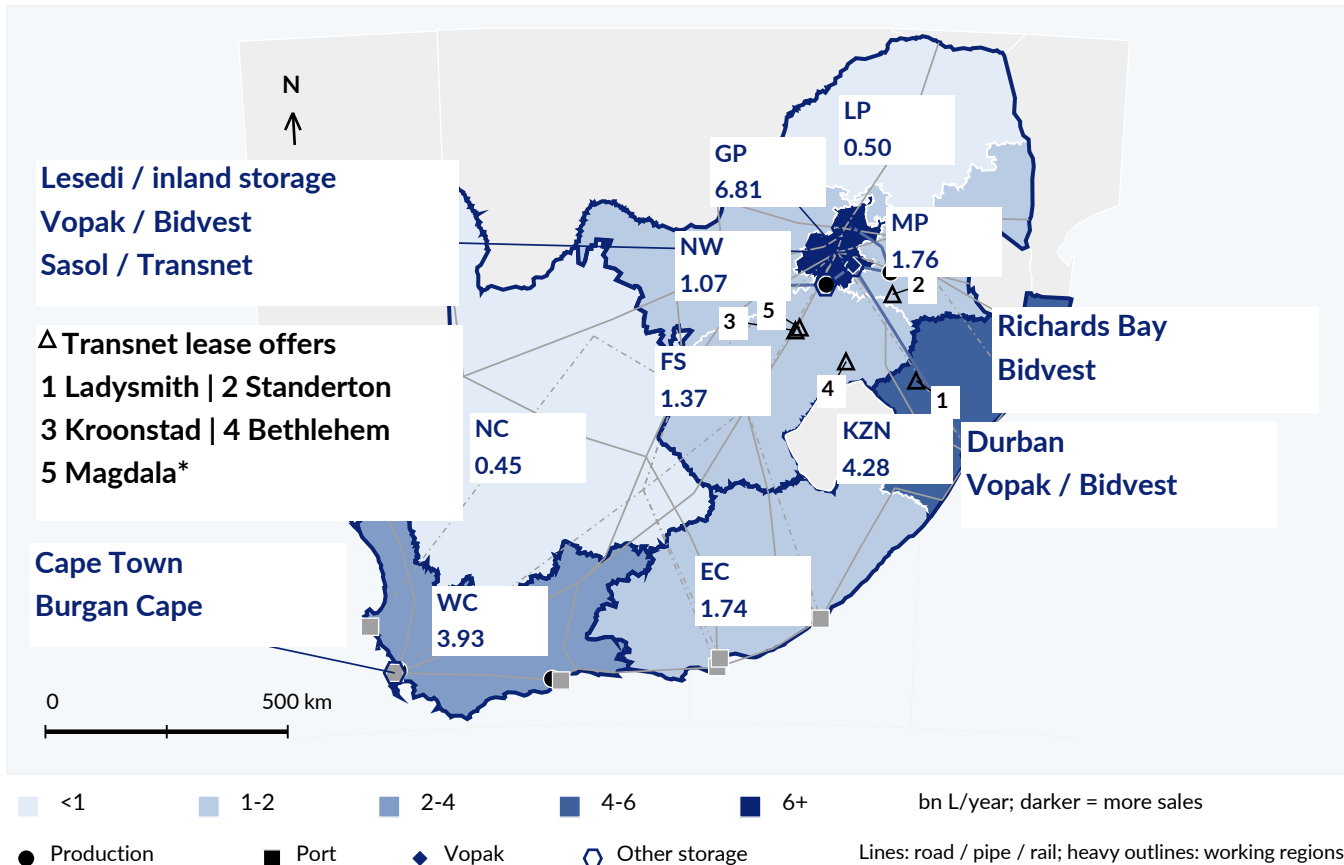
SCR reference	Evidence and required next step	Proposed lead	Timing and output	Deep dives
S1 Situation	Demand baseline and regional market Have: provincial sales history and map. Reconcile six flagged years; refresh a complete year.	Manish Nigel review	W1–W2 Sourced demand baseline	3 / 4
S2 Situation	Domestic supply, finished imports and entry ports Gap: matched-year output, trade and stocks. Reconcile the national balance before allocating provincial origin.	Manish Nigel access	W1–W3 Fuel balance + port bridge	5 / 6
C1 Complication	Demand drivers and levers Power diesel; road/rail; BEV/hybrid and efficiency; GDP/price/activity. Define baselines and alternatives, then calibrate litres effects.	Manish Nigel review	W1 define; W2–W4 test Lever-to-demand bridge	7
C2 Complication	Refinery and plant scenarios Resolve repeated refinery keys; verify closures/restarts and Sasol/Natref mechanisms. Separate actual status from conditional proposals.	Manish Henry / Nigel	W1–W3 Plant output + imports	8
C3 Complication	Economic accessibility and competing routes Cost map is illustrative. Compare Durban–NMPP–Lesedi, road/rail, Matola and Walvis Bay for the same product and destination.	Manish Nigel / Henry	W1 scope; W2–W4 test Delivered R/litre + access	9 / 10
R1 Resolution	Actual Vopak share and contestable customer demand Share is unknown. Obtain client flows, count transfers once, and test customer access against competing storage and routes.	Nigel client data Manish analysis	W1 request; W2–W4 test Served + contestable litres	11 / 12 / 13
R2 Resolution	Storage and optional investment Partial published stocks are available. Verify usable product capacity; size additional service needs only after the incremental flow case.	Nigel priority Manish / Henry	W4–W6 if prioritised Flow-to-stock / investment	14 / 15

Petrol + diesel; jet separate. Owners and weeks are proposed. Open gaps and illustrations remain explicitly labelled.

SCR S1 | Situation | Return to overview: page 2

Locate fuel demand alongside the infrastructure serving it

Petrol + diesel sales by province | 2022, bn litres/year



Evidence and implications

01 | Demand is concentrated

Gauteng, KwaZulu-Natal and Western Cape account for 68.6% of reported 2022 petrol/diesel sales.

02 | Regions are working groups

Province boundaries define reporting groups; they do not establish terminal catchments.

03 | Sales are historical demand evidence

The map is sourced provincial sales, rather than a current-year outlook or commercial accessibility result.

Next steps | proposed owners

Manish: source a later complete year and test actual destinations, route costs and customer access.

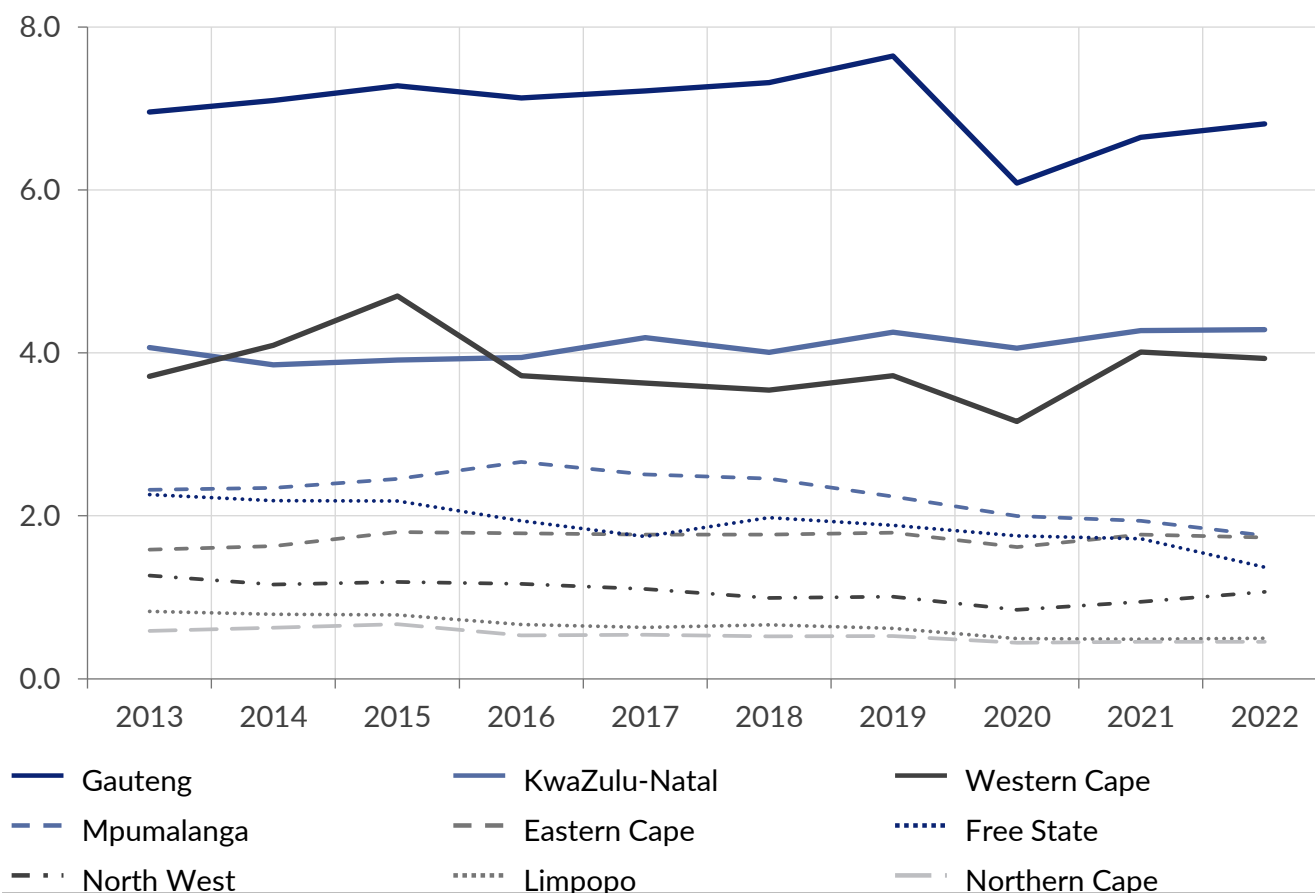
Lease offers are not operating capacity. *Magdala RFP location narrative conflicts; briefing coordinate used. Jet excluded.

Source: DMPR 2022 sales; geoBoundaries/OCHA/MDB 2020 provinces (CC BY 3.0 IGO); operator sources on slide 12.

[SCR S1 | Situation](#) | [Return to overview: page 2](#)

Track petrol and diesel sales by province over time

Annual provincial sales | 2013–2022, bn litres/year



Evidence and implications

01 | History ends in 2022

The latest complete provincial year in this extract is 2022.

02 | 2023 is incomplete

Only Q1 is present; it is excluded from annual comparisons.

03 | Six years are flagged

Provincial sums differ from national sales in 2013, 2014, 2015, 2017, 2018 and 2021.

Next steps | proposed owners

Manish: reconcile each flagged value to its original workbook cell and log old value, proposed value and reason.

SCR S2 | Situation | Return to overview: page 2

Reconcile domestic supply and finished imports

Situation | national balance and import entry ports

EVIDENCE GAP | analysis specification, not a populated result

Domestic production versus finished imports

Matched recent demand, output, imports, exports and stocks; actual provincial origin unavailable.

Output: $\text{Production} + \text{imports} - \text{exports} - \text{stock build} = \text{consumption by fuel/year}$; unit/coverage bridge and residual explained.

Import entry ports and origin

Product-specific finished imports by port/year and source country; liquid bulk is not all fuel. Output: National finished imports tied to evidenced port allocations; unallocated remainder explicit.

Evidence and implications

01 | Evidence status

These topics remain open. Existing calculations or public source leads do not establish calibration or closure.

02 | What is already available

Energy balance 2007-2021 staged; provisional supply engine; p6 provincial source illustration.

03 | What this page will resolve

A traceable petrol/diesel result with source, period, units and explicit unassessed residual. Jet stays separate.

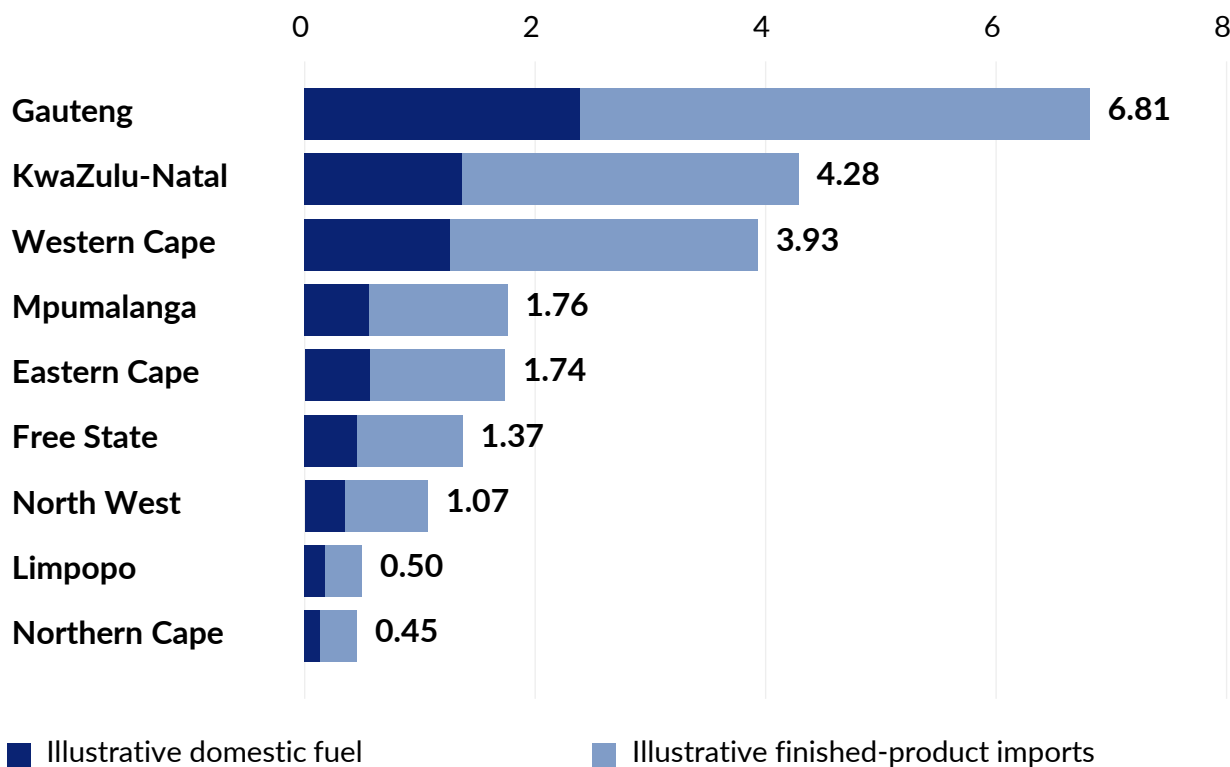
Next steps | proposed owners

Manish: reconcile the national fuel balance, then bridge finished imports to entry ports. Nigel: secure source access. Explain all unit conversions and unallocated volumes.

[SCR S2 | Situation](#) | [Return to overview: page 2](#)

Shape the provincial split between domestic fuel and imports

2022 sales totals | ILLUSTRATIVE domestic/import allocation



Units: bn litres/year. Imported/domestic segments are a scenario illustration, not measured provincial flows.

Evidence and implications

01 | Demand totals are sourced

The provincial bars use reported 2022 petrol/diesel sales.

02 | Origin shares are illustrative

Domestic/import fractions are authored examples; no actual provincial origin allocation is established.

03 | Imports need a national balance

Exports and stock movements must be reconciled before a sourced import requirement can be presented.

Next steps | proposed owners

Manish: populate the matched-year national balance first; retain provincial origin as unknown until supported.

SCR C1 | Complication | Return to overview: page 2

Translate demand drivers into explicit fuel levers

Complication | power, freight, fleet and economic activity

EVIDENCE GAP | analysis specification, not a populated result

Power-generation diesel and potential reversal

Historical GWh-to-litres tie; baseline/alternative dispatch, units, dates and incremental demand effect.

Road-to-rail freight shift

Specific intervention with baseline/alternative mode shares, dates and diesel-intensity bridge; no duplicate mileage reduction.

BEV, hybrid, efficiency and charging effects

Opening-fleet/cohort tie; drivetrain-specific L/100km; dated uptake alternatives and litres effect.

GDP, fuel prices and kilometres travelled

Explicit historical/forecast boundary and consuming function; calibrated sensitivity without duplicate activity reductions.

Evidence and implications

01 | Evidence status

These topics remain open. Existing calculations or public source leads do not establish calibration or closure.

02 | What is already available

Eskom OCGT observations 2022-2026 staged; engine capacity/load factor/efficiency path.

03 | What this page will resolve

A traceable petrol/diesel result with source, period, units and explicit unassessed residual. Jet stays separate.

Next steps | proposed owners

Manish: define baseline, alternatives, units, dates and consuming equation for each lever. Nigel reviews; calibrate against source observations.

Test refinery availability and plant-specific scenarios

Complication | refinery closures, restart proposals and Sasol / Natref

EVIDENCE GAP | analysis specification, not a populated result

Refinery closures, availability and SAPREF proposal

Current output/yields/status; dated closures/restarts; SAPREF project evidence; 558 extra refinery rows unresolved.

Output: Unique country/year/scenario keys with source cells; actual/proposed distinction; per-refinery output tie and import sensitivity.

Secunda gas/MRG and Natref future

MRG timing and causal liquid-fuel effect; Natref future not established; partner propositions untested.

Output: Documented plant-level mechanism and dated scenario; no gas-to-liquid displacement inferred without evidence.

Evidence and implications

01 | Evidence status

These topics remain open. Existing calculations or public source leads do not establish calibration or closure.

02 | What is already available

Refinery assumptions and utilisation 2008-2050; provisional supply engine; duplicate-row diagnostics.

03 | What this page will resolve

A traceable petrol/diesel result with source, period, units and explicit unassessed residual. Jet stays separate.

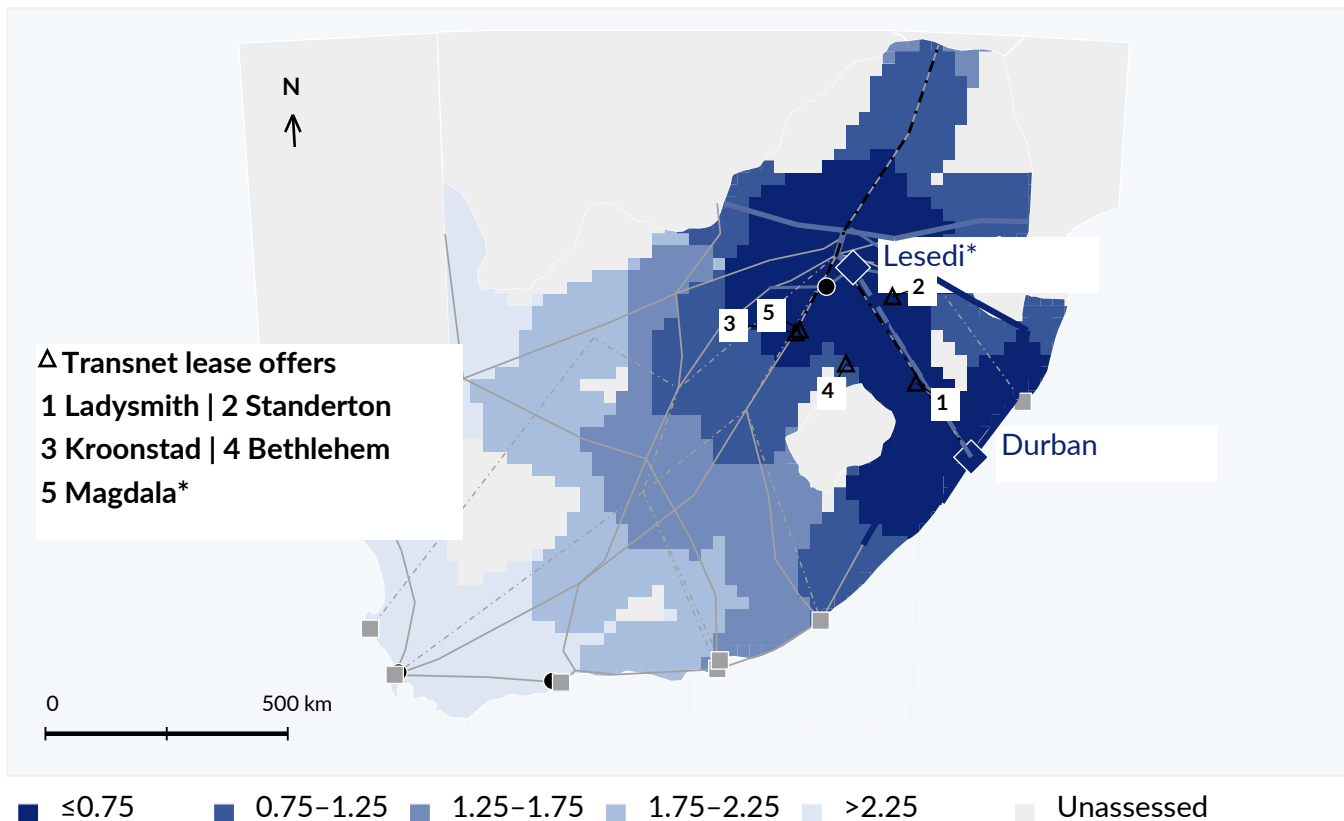
Next steps | proposed owners

Manish: resolve duplicate refinery keys and source plant status. Henry: review the gas/liquid mechanism. Nigel: agree conditional refinery scenarios.

SCR C3 | Complication | Return to overview: page 2

Map conditional market access from Durban and Lesedi

Road-delivery accessibility | illustrative transport cost, R/litre



Example rate: R0.15/L dispatch + R0.002/L/km. Grey = unassessed.

Illustrative road cost only. Pipeline/rail access and commercial rights are not priced.

Source: Existing appendix routes; illustrative candidate connections, not verified catchments. Natural Earth 1:50m.

Evidence and implications

01 | Darker cells mean lower example cost

The surface shows illustrative delivered road-transport cost in R/litre.

02 | Other modes remain context

Pipeline and rail costs, capacity and access are not evaluated in this surface.

03 | Physical proximity is insufficient

No operational or commercial catchment is validated by this illustration.

Next steps | proposed owners

Manish and Nigel: select the first destination and obtain dated route tariffs, quotations and access evidence.

SCR C3 | Complication | Return to overview: page 2

Compare routes on the same delivered-cost basis

Complication | Durban-NMPP-Lesedi, road / rail, Matola and Walvis Bay

EVIDENCE GAP | analysis specification, not a populated result

Durban-NMPP-Lesedi versus road/rail, Matola and Walvis Bay

Comparable route costs, NMPP capacity/tariff, actual rail service and cross-border feasibility.
Output: Comparable delivered R/litre with cost components, capacity/access limits and dated sources; no national optimisation claim.

Evidence required before quantification

Schematic appendix routes and illustrative Durban/Lesedi road-cost map.
No operational route allocation or delivered-cost engine.

Evidence and implications

01 | Evidence status

These topics remain open. Existing calculations or public source leads do not establish calibration or closure.

02 | What is already available

Schematic appendix routes and illustrative Durban/Lesedi road-cost map.

03 | What this page will resolve

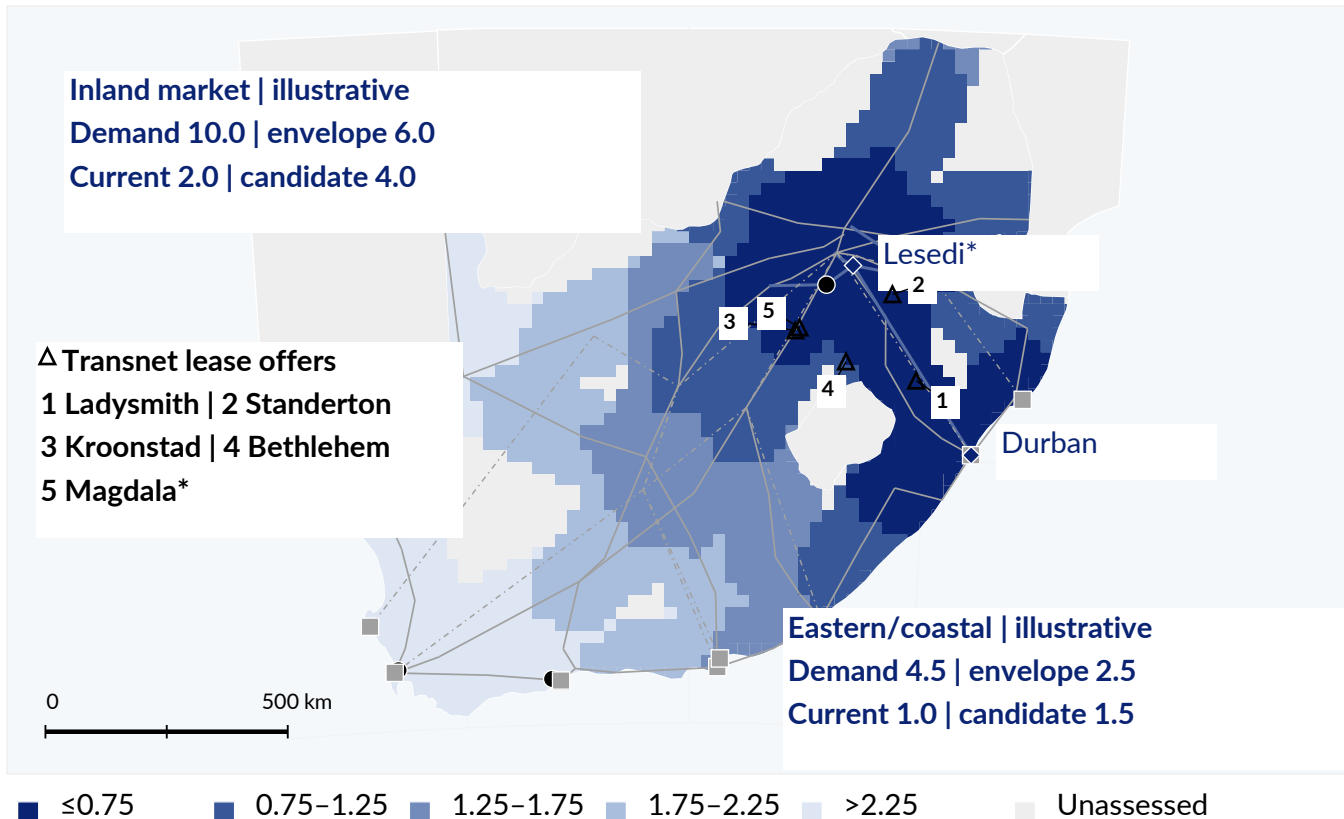
A traceable petrol/diesel result with source, period, units and explicit unassessed residual. Jet stays separate.

Next steps | proposed owners

Nigel: choose the first product and customer destination. Manish: compare dated route costs, capacity and access on the same R/litre basis. Henry reviews.

Separate demand, accessible volume and Vopak flows

Illustrative road cost (R/litre) and market volumes



Receipts 2.8 + 2.0 – transfer 1.8 = unique demand 3.0

Example rates: R0.15/L + R0.002/L/km. Grey = unassessed. Volumes: bn L/year.

Source: Illustrative CSVs and road transport rates; no actual Vopak throughput or quoted transport costs supplied.

Evidence and implications

01 | Balance arithmetic is illustrative

Example demand 21.0 less domestic production 7.0 gives imports 14.0, before exports and stocks.

02 | Transfers count once

Example Durban receipts 2.8 plus Lesedi receipts 2.0 less shared transfer 1.8 gives unique served demand 3.0.

03 | Candidate volume is conditional

The example envelope is 8.5 and additional candidate volume 5.5; neither is forecast capture.

Next steps | proposed owners

Nigel: request client flows. Manish: replace examples with matched-year product and destination records.

SCR R1 | Resolution | Return to overview: page 2

Size each regional market and locate competing storage

All four regions | reported 2022 demand, bn litres/year

National total 21.90 bn L | petrol + diesel; jet excluded

Region / provinces	Reported demand	Vopak served	Additional contestable
	0 4 8 12		
Eastern coast EC/KZN	6.02	Not established	Not established
Inland GP/FS/LP/MP/NW	11.50	Not established	Not established
Western coast WC	3.93	Not established	Not established
Other Northern Cape	0.45	Not established	Not established

To calculate share: unique Vopak customer deliveries / same-year regional demand.

Contestable demand needs route cost, compatible tanks, contracts and switching evidence.
Count Durban-Lesedi transfers once. Region boundaries are working groupings, not catchments.

[1] Vopak [2] Bidvest [3] Borgan [4] Shell [5] NERSA [6] Sasol [7] Transnet

Source: DMPR 2022 provincial petrol/diesel sales; public operator sources [1-7], checked 6 Oct 2026. Share not established.

Regional competitor evidence

01 Eastern coast

Vopak: Durban [1]. Bidvest: Durban and Richards Bay [2]; mixed-product capacity.
Transnet: Ladysmith lease tanks [7].

02 Inland

Vopak: Lesedi [1]. Bidvest: Isando [2].
Sasol: Alrode, Pretoria West, Waltloo, Sasolburg [6].
Transnet: Tarlton / Jameson Park + four lease sites [7].

03 Western coast

Borgan Cape: Cape Town petrol/diesel terminal [3].
Confirm current tank availability and customer access.

04 Other / Northern Cape

Operator coverage incomplete. Check Shell site notices [4] and NERSA licences/access records [5].

SCR R1 | Resolution | Return to overview: page 2

Establish actual share and additional contestable demand

Resolution | unique deliveries, reachable customers and commercial access

EVIDENCE GAP | analysis specification, not a populated result

Vopak market share at Durban and Lesedi

Unique customer deliveries, receipts, shared transfers, destinations and matched period.

Output: Separate terminal throughput and unique served-demand shares; matched-year/product denominator; traceable records, no capacity-as-share.

Additional commercially contestable market

Usable product-compatible tanks, spare routes, contracts, switching, customer overlap and Northern Cape inventory.

Output: Accessible demand by product/region/period; explicit assumptions and unassessed residual; competitors volume is not automatically capturable.

Evidence and implications

01 | Evidence status

These topics remain open. Existing calculations or public source leads do not establish calibration or closure.

02 | What is already available

Sourced regional sales p12; published capacity p14; illustrative transfer arithmetic p11.

03 | What this page will resolve

A traceable petrol/diesel result with source, period, units and explicit unassessed residual. Jet stays separate.

Next steps | proposed owners

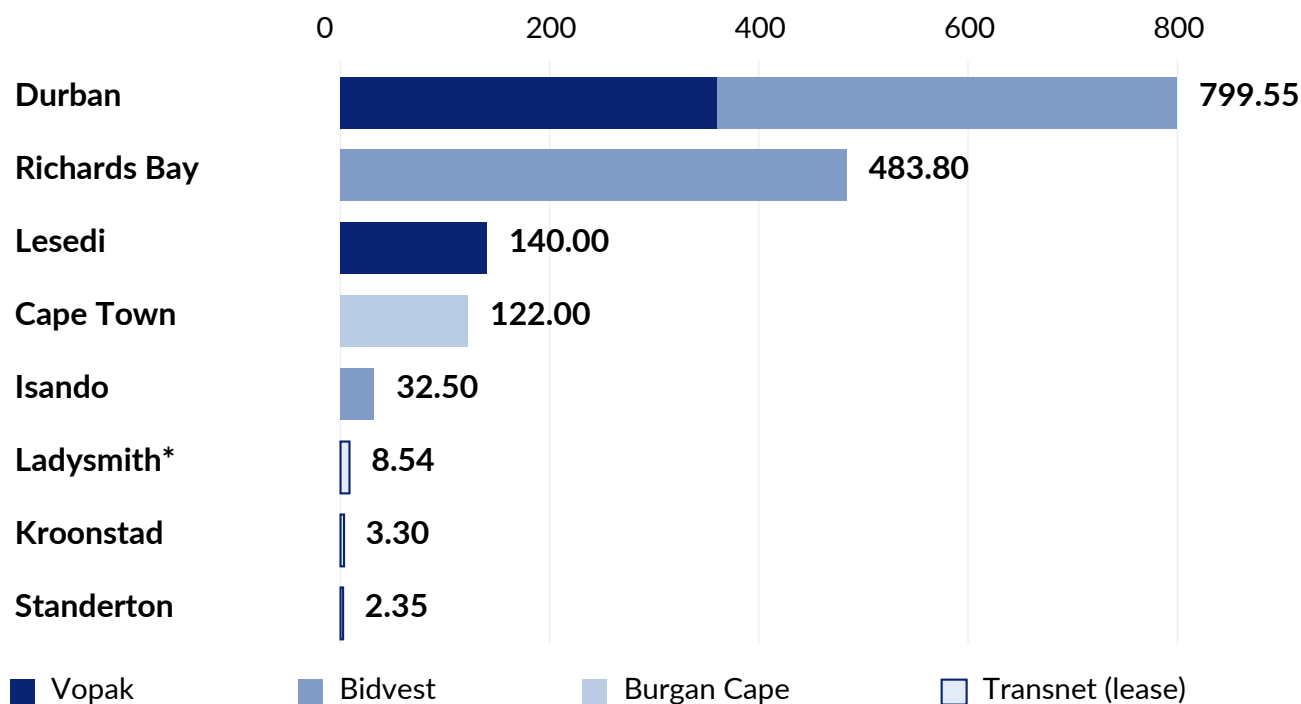
Nigel: request client flow and destination records.
Manish: reconcile transfers, calculate unique served demand, then test additional customer access and competing storage.

[SCR R2 | Resolution](#) | [Return to overview: page 2](#)

Compare published storage capacity by location

Published storage by location | thousand m³; mixed capacity bases

One shared scale for every location



Small lease bars reflect smaller tank volumes; they are not operating supply.

*Ladysmith excludes 0.474 intermixture. Unquantified: Bethlehem, Magdala, Sasol, Tarlton, Jameson Park.

Source: Vopak; Bidvest; VTTI Borgan Cape; Transnet lease RFPs (5 Oct 2026), pp. 2-3. Checked 6 Oct 2026.

Evidence and implications

01 | Capacity is a stock

Published tank volumes are not annual fuel deliveries or market share.

02 | The inventory is partial

Missing capacities are unknown, rather than zero; mixed product scopes limit comparison.

03 | Leases are conditional

Transnet lease tanks are not counted as currently available operating supply.

Next steps | proposed owners

Manish: verify product compatibility, capacity basis and operating status. Nigel: confirm usable client capacity.

SCR R2 | Resolution | Return to overview: page 2

Gate investment on evidenced incremental customer flows

Resolution | optional service and investment assessment

EVIDENCE GAP | analysis specification, not a populated result

New storage and investment scope

Incremental accessible flows, turnover/peak inventory, product compatibility, cost and commercial return criteria.

Output: Throughput-to-usable-stock bridge with seasonality, inventory days and dispatch limits; downside sensitivity; no investment claim from import deficit alone.

Evidence required before quantification

Partial gross/lease stocks, without client handling/inventory profile.

Detailed service sizing deferred; infrastructure secondary to fuel integration.

Evidence and implications

01 | Evidence status

These topics remain open. Existing calculations or public source leads do not establish calibration or closure.

02 | What is already available

Partial gross/lease stocks, without client handling/inventory profile.

03 | What this page will resolve

A traceable petrol/diesel result with source, period, units and explicit unassessed residual. Jet stays separate.

Next steps | proposed owners

Nigel: decide whether to prioritise investment analysis after the flow case. Manish and Henry: bridge incremental flows to usable stock and service constraints.

Agree priorities and next steps

Confirm demand, routes and access

6 October 2026

Internal working discussion

Photo source: Royal Vopak | Terminal Lesedi | vopak.com

